

(주)동인엔시스

COMPANY PROFILE 2026

Controlling equipment, connecting data.

Smart manufacturing solutions since 1991 · Intelligent Control Panel

donginensis.com

After 35 years of building control panels, we are turning them into data infrastructure.

Since 1991, DONG-IN ENSIS has designed and built control panels for industrial and marine sites. On that foundation, we now make Intelligent Control Panels that collect and connect equipment data while controlling it.



Intelligent Control Panels on a factory floor · illustration

35⁺

Years

Founded in 1991 as a supplier of industrial control equipment

750⁺

Projects

Marine, energy and plant control system projects

15⁺

Industries

Shipbuilding, hydrogen, energy, manufacturing, data centers

40⁺

CE·UL

CE·UL certified solutions

Industrial DX and AX start with field data.



Thank you for visiting DONG-IN ENSIS.

We began in 1991 as a supplier of industrial control equipment and grew into a company that designs and builds control systems for ships, energy facilities and plants. Over more than 750 projects, one thing became clear:

most equipment is already controlled by PLCs and drives, but the data they generate never leaves the control panel.

AI and smart factories need field data first. Our Intelligent Control Panel collects data where the equipment is controlled and delivers it to the cloud or customer servers in a standard format. For manufacturers, it is the first step toward managing equipment with data. For IT companies, it is a reliable source of field data.

We will keep adding data and intelligence to the control technology we have built in the field for 35 years.

CEO, DONG-IN ENSIS CO., LTD. **Min-gi Baek**

Company overview

Company	DONG-IN ENSIS CO., LTD. (주)동인엔시스	Slogan Controlling equipment, connecting data. 설비를 제어하고, 데이터를 연결하다.
CEO	Min-gi Baek	
Founded	October 1991 Incorporated July 1997 · renamed 2023	
Business	Intelligent Control Panels · industrial and marine control systems System integration · supply of global power and control products	
Sites	Head office, Busanjin-gu, Busan, Korea Eco-Technology Research Center (NS Tower) · Yangsan Factories 1 & 2 · Gijang Factory	
Management	ISO 9001 · ISO 14001 · ISO 45001	
Designations	Venture Company · INNOBIZ · MAINBIZ · Corporate R&D Center · Smart Factory Supplier Hydrogen Specialized Company · Innovation Premier 1000 (2026) · National Innovative Company 1000 Busan Strategic Industry Leading Company · Legend 50+ · Busan Service Strong SME	
Partners	Schneider Electric official SI partner Supply partner for ABB · Fuji Electric · Eaton · Danfoss	

VISION

Accelerating **DX and AX** in manufacturing and maritime industries with field equipment data.

MISSION

Creating and connecting **reliable data** from the control panels that run the equipment.



Intelligent Control Panel at a hydrogen facility · illustration

From components to systems, from systems to data



Three foundations behind the Intelligent Control Panel

Data infrastructure is not built with software alone. It starts with hands-on experience controlling real equipment.

01 · MANUFACTURING

We design and build in-house

From panel design to build, wiring, PLC programming, factory testing and on-site commissioning.

- Yangsan Factories 1 & 2 (2023) · Gijang Factory
- Dedicated FAT test line
- Design and build to CE·UL standards

02 · FIELD EXPERIENCE

We control demanding sites

Control systems built and verified at hydrogen, shipbuilding and plant sites with strict safety requirements.

- Liquid hydrogen station control nationwide
- FGSS test facility control for a major shipyard
- Hybrid electric propulsion for green ships

03 · R&D

We research data and predictive control

Our Eco-Technology Research Center leads national and regional R&D and industry-academia work.

- 5 registered patents on LH2 station control
- National R&D on hydrogen-powered vessels
- Joint research with Korea Maritime & Ocean Univ.

Global brand partners

Schneider Electric official SI partner

ABB

Fuji Electric

Eaton

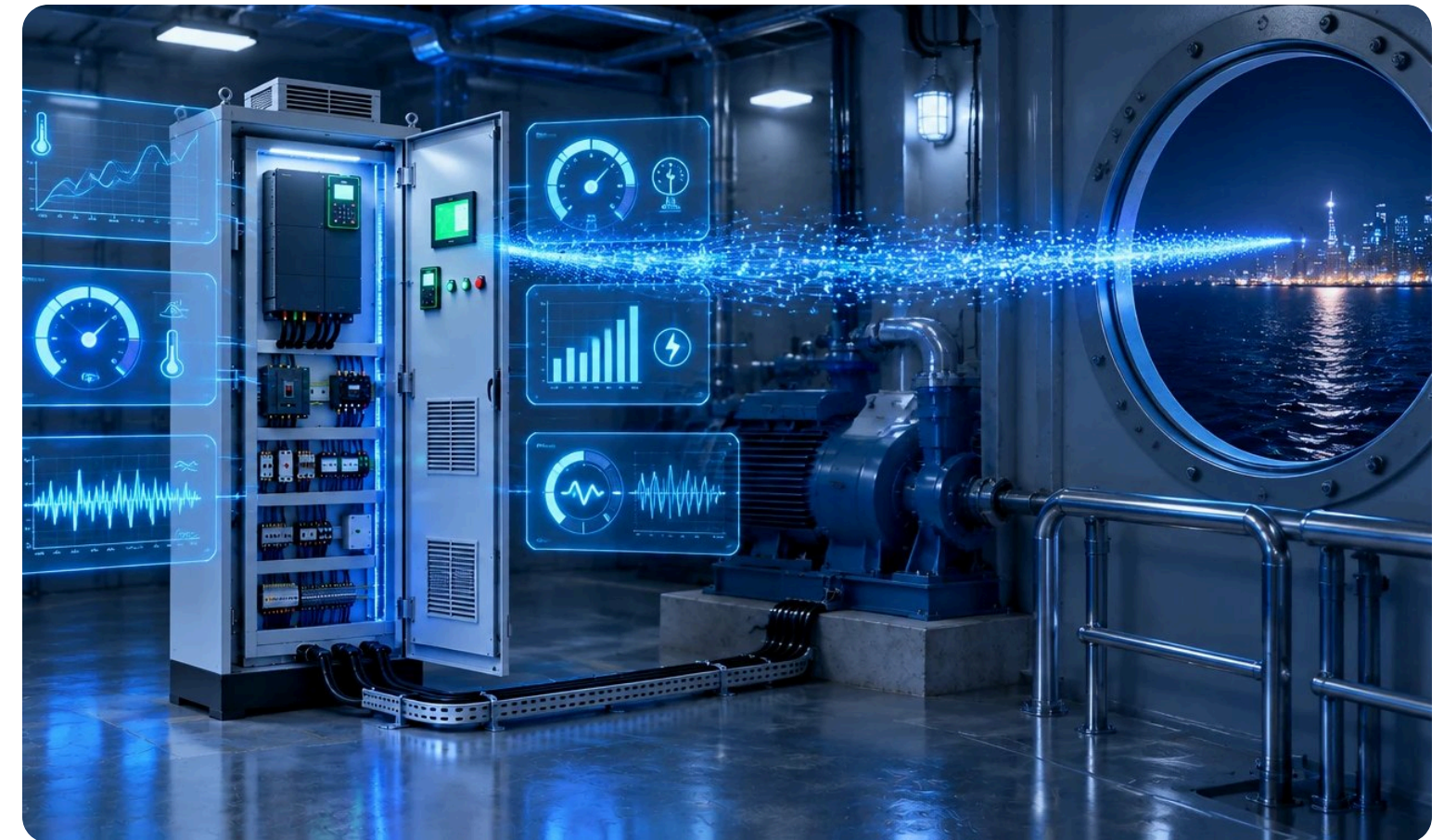
Danfoss

Intelligent Control Panel

From equipment control to data connectivity, one Intelligent Control Panel.

It controls equipment with a PLC while collecting operating data, and connects securely to the cloud or customer servers without extra data collection devices.

Motor winding temp. U·V·W Bearing temp. Speed Voltage · current · energy
Vibration Ambient temp. · humidity Run status



Intelligent Control Panel collecting and connecting equipment data · illustration

01

Real-time PLC control & data collection

Control logic and interlocks stay with the field PLC.

02

MQTT / TLS transmission

The PLC sends operating data over MQTT, protected by TLS.

03

DataHub standard time series

Equipment data is organized in a common structure and stored over time.

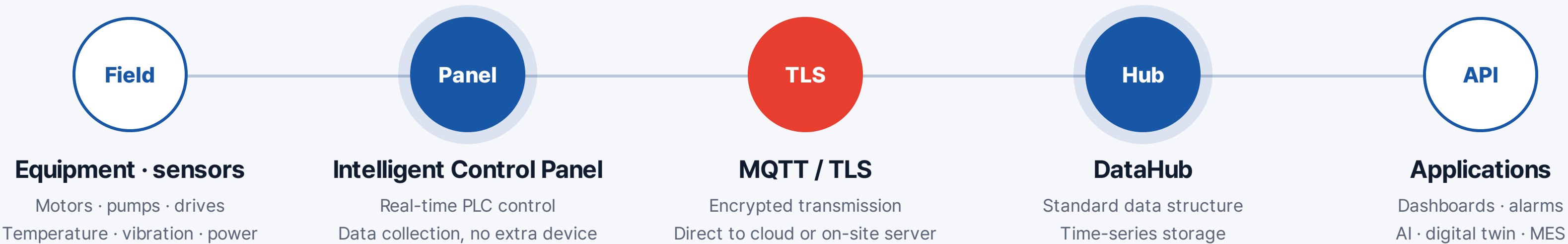
04

REST API integration

REST API and WebSocket for MES, monitoring and analytics systems.

How It Works

From field equipment to AI, seamlessly connected



Cloud

Connect to the DONG-IN ENSIS cloud

- Fast deployment with no server to set up
- Check anytime on web and mobile
- Manage multiple plants and sites on one screen

On-premise

Install on customer servers and networks

- DataHub and dashboard run in-house, without internet access
- Use the customer's own server or one supplied by us
- For sites whose security policy rules out external cloud

Both options use the same data structure and API. Maintenance for on-premise systems is agreed in line with the customer's security policy.

Equipment safety and data security, with no extra collection devices

01 Control stays in the field

Interlocks and final control remain with the field PLC. Servers never control equipment directly.

02 Encrypted transmission

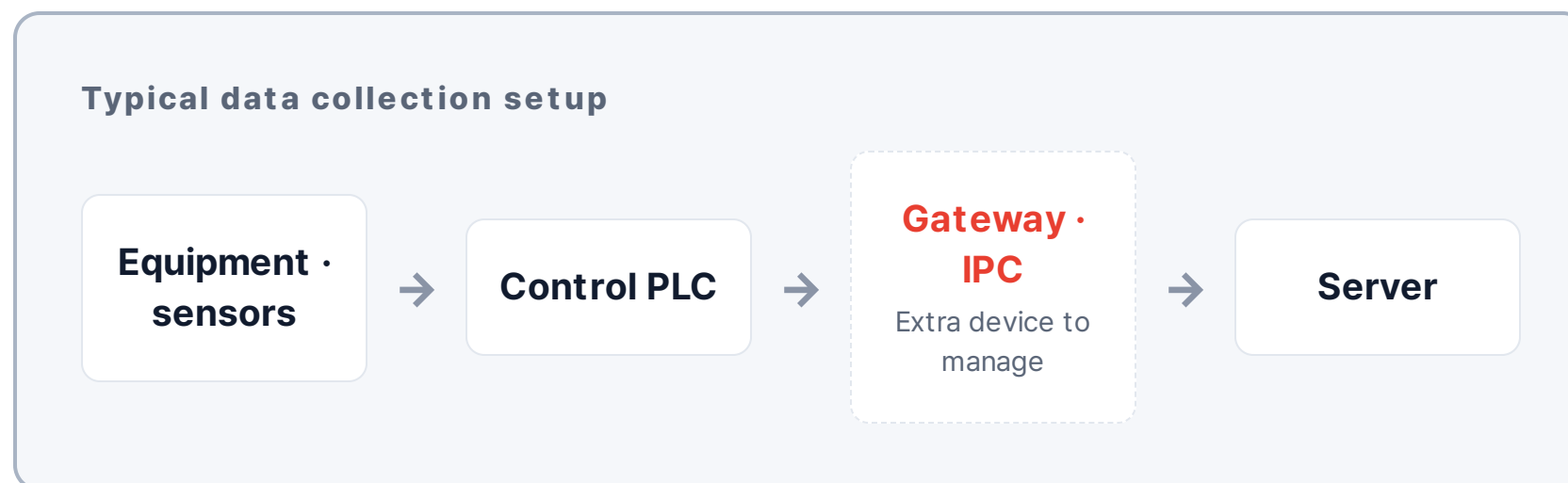
MQTT traffic between the site and the cloud (or customer server) is encrypted with TLS.

03 Runs without external networks

On-premise systems run without internet access; cloud systems can connect over LTE, separate from the plant network.

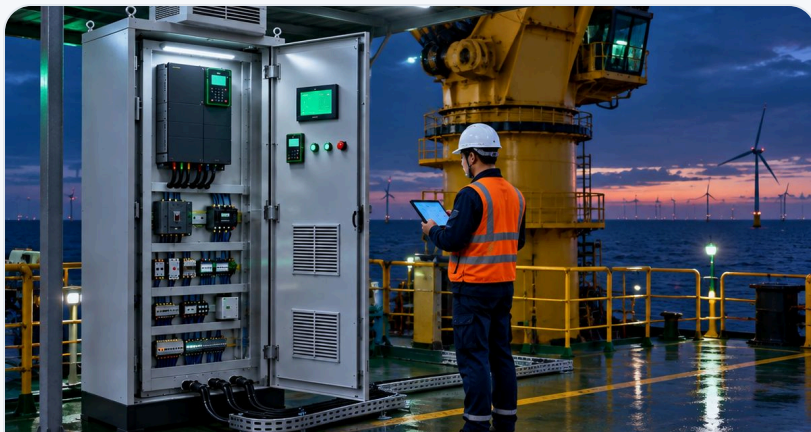
04 Access and data use

Accounts, permissions, scope of data use and retention periods are set in each project contract.



* Network and security configuration is agreed according to site conditions and the customer's security policy.

From the sea to data centers, wherever motors run



MARINE · OFFSHORE

Shipbuilding · marine

Control proven on a moving sea

- LNG carrier engine rooms
- Ballast water treatment
- Wind installation vessels · port cranes



HYDROGEN · ENERGY

Hydrogen · energy

Hydrogen control backed by patents

- Liquid hydrogen stations
- Hydrogen-powered vessels
- 5 patents · Hydrogen Specialized Co.



MANUFACTURING

Smart factory

Start a smart factory with a panel swap

- Steel blowers · air compressors
- Water pumping stations
- Production lines · food & pharma



DATA CENTER

Data centers

Nonstop cooling, lower power

- Chillers · cooling water pumps
- Cooling tower fans · liquid cooling CDU
- DCIM · BMS integration

Water booster pump
station
160kW × 3

Steel mill blower
250kW

Air compressor room
90kW × 2

Liquid hydrogen
station
30kW

LNG carrier engine
room
75kW × 2

Ballast water
treatment
55kW

Wind installation
vessel crane
200kW

Port container crane
300kW

* Application scenarios are examples. Images are illustrations.

Green ships and hydrogen facilities, proven at the most demanding sites



HYDROGEN VESSEL

Hydrogen-powered vessels

- Ministry of Oceans and Fisheries project on small hydrogen-powered vessel technology and demonstration
- Integrated control of fuel cells, propulsion motors and hydrogen tanks
- Predictive power control and output distribution



HYBRID PROPULSION

Hybrid electric vessels

- Hybrid propulsion and integrated control for the Chungnam Hospital Ship and Gyeongnam Cheongjeong (2023)
- Diesel-electric integrated control for a pilot hybrid fishing vessel (2026)
- National R&D on hybrid fishing vessel equipment (2021)



LIQUID HYDROGEN

Liquid hydrogen stations

- Control systems for a liquid hydrogen station project led by a major Korean company
- Field-proven at commercial stations
- 5 control system patents · Hydrogen Specialized Company

* Images are illustrations.

Hybrid propulsion for green ships, with our integrated control & monitoring



Chungnam Hospital Ship · G/T 330 t

- Diesel engine + electric motor hybrid propulsion
- Propulsion system · integrated control & monitoring
- In service Aug 2023 · green-ship commendation



Gyeongnam Cheongjeong · G/T 120 t environmental cleanup ship

- Diesel engine + electric motor hybrid propulsion
- Propulsion system · integrated control & monitoring
- In service Apr 2023



Harbor cleaning ship · G/T 280 t

- Diesel · LNG engine hybrid propulsion
- Fuel gas supply system (FGSS) control system
- Gas safety control on an LNG-fuelled vessel

* Pilot hybrid fishing vessel: diesel-electric integrated control (2026) · National R&D: hybrid electric fishing vessels (2021-25), safety-based small hydrogen-powered vessels (2022-26)

Major Reference · LNG Fuel Gas Supply System

LNG fuel gas supply system (FGSS) control, delivered again and again for newbuilds

Shipyard · owner	Vessel · facility	Flag · class	Scope of supply
Jiangsu New Times Shipbuilding (NTS)	EPS-NTS 210,000 / 209,000 DWT bulk carriers, 13 ships	Liberia · ABS	FGSS glycol water pump control · hydraulic power supply system
Namura Shipbuilding	95,000 DWT bulk carrier (S496)	Liberia · NK	FGSS glycol water pump control system
GSI (Guangzhou Shipyard International)	HMM 8,600 CEU DF car carriers, 3 ships	Panama · KR	FGSS glycol pump · L.O pump & oil heater control
GSI (Guangzhou Shipyard International)	EPS/Chandris 111K DWT LNG DF crude/product tanker	Taiwan · KR	BOG compressor · glycol pump · L.O pump & oil heater control
Samsung Heavy Industries	FGSS system test facility	—	G/W (glycol water) heater control system

13 ships

NTS 210K · 209K DWT bulk carriers

ABS · NK · KR

Class of FGSS vessels supplied

China · Japan

NTS · Namura · GSI shipyards



8,600 CEU DF car carrier · 111K DWT LNG DF tanker

* Ship photos show the vessel types. We are extending our gas supply and safety control experience on LNG-fuelled ships to hydrogen facilities.

Hydrogen stations, power plants and test facilities: control systems for energy facilities



2024 · Liquid hydrogen



SK E&S liquid hydrogen station

- VFD · PLC · UPS control system (2024)
- Proven in commercial operation
- 5 patents on hydrogen control & power



Fuel cell plant CCUS BOP control

- KOSPO fuel cell power plant
- BOP control for carbon capture (CCUS) facilities
- SK ecoplant



KMOU · Korean Register · KOMERI

MASTC marine test facility

- Propulsion test facility for green ships
- 400 kW propulsion drive · main switchboard · PMS
- Load bank · battery charge/discharge (DC/DC) · HiL

From overseas production lines to compressor OEMs, industrial plant control references



PLANT · USA

**Hyundai Motor Group
Metaplant America**

- EV production line in the USA
- Integrated control systems built and supplied



PLANT · 7 COUNTRIES

KOS · Kiswire

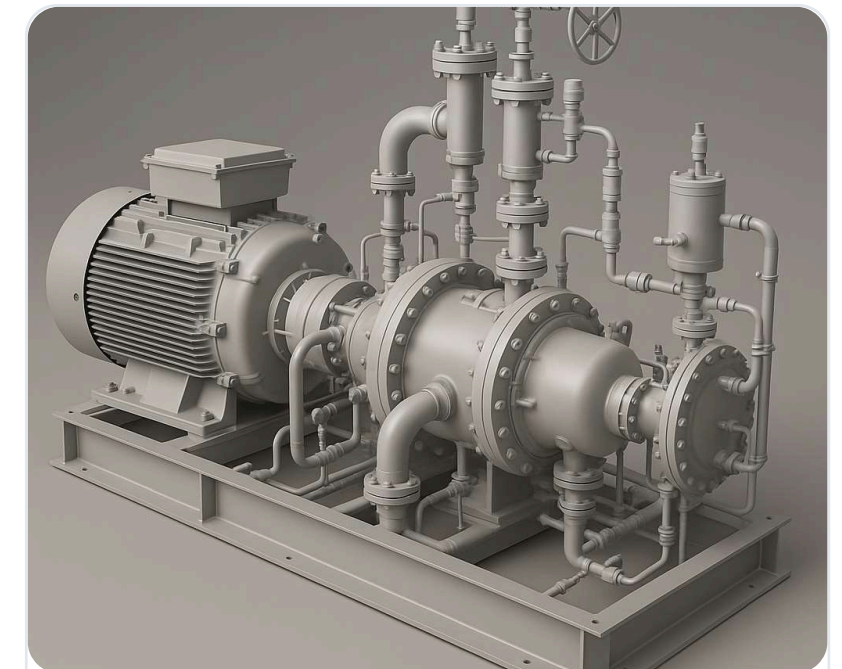
- Wire drawing · stranding · plating · heat treatment lines
- Korea · Vietnam · USA · Czechia · Hungary · Malaysia · China



OEM · COMPRESSOR

Atlas Copco

- High-performance BOG compressors for the LNG value chain
- Control system engineering · global support



OEM · COMPRESSOR

SB Sunbo

- BOG compressors for LNG carriers
- Control system design · supply · commissioning

Multi-axis servo and motion control, control systems for special machines



PACKAGING

BOBST Korea

- Packaging machinery
- Special machine control engineering · supply

Servo · motion control



TEXTILE

Daehan Synthetic Fiber auto doffer

- Auto doffer for the textile process
- PLC · HMI control system

M340 PLC · 8 servo axes (CANopen)
2 × ATV312 drives · HMI



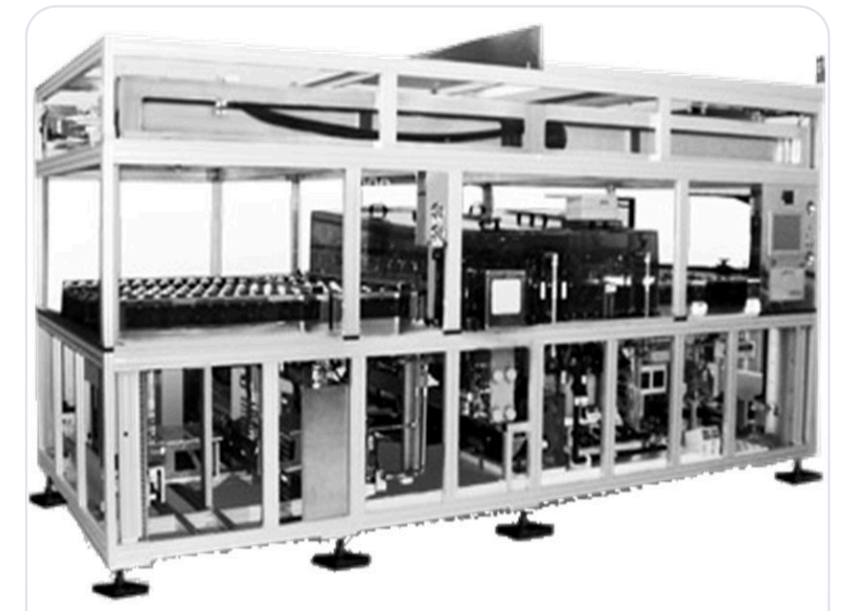
SMR

NUCLEAR

Power MnC SMR refueling machine

- Refueling machine for small modular reactors
- PLC · monitoring PC control system

M580 PLC · 3 servo axes
ATV930 drive · monitoring PC



CLEANER

DMS high-density cleaner

- High Density Cleaner
- Distributed control · HMI

5 distributed PLCs · servo (CANopen)
Modbus TCP · HMI

※ IGis thermal plastic spacer production line: M262 motion controller · 13 servo axes (Sercos III) · 2 drive axes · Schneider Electric platform

Reference List

Major references

Field	Owner · customer	Vessel · facility	Scope of supply
Green ships	Chungcheongnam-do	Chungnam Hospital Ship · G/T 330 t	Hybrid propulsion · integrated control & monitoring
	Gyeongsangnam-do	Gyeongnam Cheongjeong · G/T 120 t	Hybrid propulsion · integrated control & monitoring
	MOF Yeosu Regional Office	Harbor cleaning ship · G/T 280 t	Diesel-LNG hybrid · FGSS control
LNG FGSS	NTS · Namura · GSI	Bulk carriers · car carriers · LNG DF tanker	FGSS glycol pump · BOG · L.O pump & oil heater control
	Samsung Heavy Industries	FGSS system test facility	G/W heater control system
Compressor OEM	Atlas Copco · SB Sunbo	LNG BOG compressors	Control system engineering · supply · commissioning
Hydrogen · energy	SK E&S	Liquid hydrogen refueling station	VFD · PLC · UPS control system
	SK ecoplant	KOSPO fuel cell power plant CCUS	BOP control system
	MASTC (KMOU · KR · KOMERI)	Green ship propulsion test facility	Propulsion motor drive · main switchboard · PMS · HiL
	Hyundai Motor Group	Metaplant America (USA) EV production line	Integrated control system
Industrial plants	KOS · Kiswire	Drawing · stranding · plating · heat treatment lines (7 countries)	Control systems
	Dongwha	Formalin resin plant	Entire plant control system, designed and built
	BOBST Korea	Packaging machinery	Special machine control system
Special machines	Daehan Synthetic Fiber · DMS · IGis	Auto doffer · high-density cleaner · thermal spacer line	PLC · up to 13 servo axes · drives · HMI
	Power MnC	SMR refueling machine	PLC · 3 servo axes · drive · monitoring PC
Marine panels	KSB	Marine equipment	Control consoles · system panels
National R&D	MOF · KIMST	Hybrid electric fishing vessels · small hydrogen-powered vessels	Propulsion motor drive · integrated control & monitoring

Customers

Building sites together with customers in Korea and abroad



750⁺

Control system projects

Marine · energy · plant
special machines

7 countries

Overseas production lines

Korea · Vietnam · USA · Czechia
Hungary · Malaysia · China

6 classes

Class-approved vessels

ABS · NK · KR · DNV · LR · BV
Newbuilds in China · Japan

2026

Innovation Premier 1000

Selected by MOTIE

**Hydrogen
Specialized Company**

Designated by MOTIE

Hydrogen control · power tech
5 registered patents

Smart Factory

Registered supplier

Linked to smart factory
programs
Proposal · application support

Prepare for equipment failure **with data**, not guesswork.

ISSUE 01

A machine stops at dawn and no one knows why

Crews arrive only after the failure, and lost production grows with every hour of downtime.

ISSUE 02

When experts leave, their know-how leaves too

Judgments made by sound and feel are never recorded.

ISSUE 03

Signs of overheating and overload go unrecorded

Without a history of temperature, current and vibration, prevention is hard.



Checking equipment condition with data · illustration

01 · BUILD

Intelligent Control Panel

- New installation or panel replacement
- PLC programming · FAT · commissioning
- Start with 1-2 machines, then expand

02 · DATA

Data service

- Stored in the cloud or on customer servers
- Dashboards · early-warning alarms
- MES · analytics integration (API)

03 · OPERATE

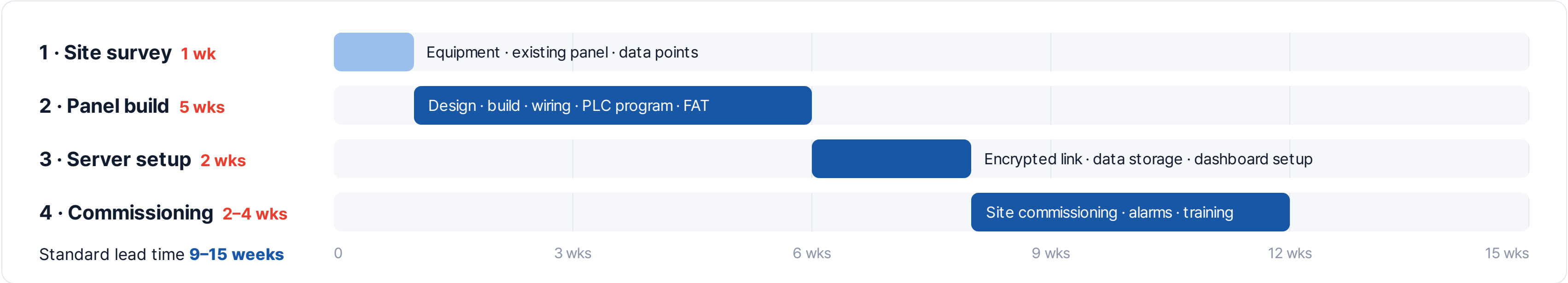
Maintenance

- Remote status checks · regular inspection
- Alarm tuning · operator training
- On-site support · parts replacement

Cost depends on the equipment and motor ratings, data points, network environment and whether an existing panel is replaced. We provide a configuration and quote after a site survey.

Process

From site survey to after-sales, one company all the way



May vary with site conditions. Panels are built and tested in our factory before installation, and replacement work is scheduled around your production plan.

Government support

Registered smart factory supplier (Korea)

1 Check programs

We review available smart factory support programs with you

2 Proposal · quote

Configuration tailored to your project plan

3 Application

Support with the technical documents required

4 Build

Build · commissioning · after-sales once selected

Roadmap

Intelligent Control Panel · data · AI

2026 · NOW

Data infrastructure

PLC → DataHub → dashboard, AI anomaly detection PoC

2027

Predictive maintenance

Anomaly detection on real data, digital twin integration

NEXT

Smart factory expansion

Integrated control of field equipment such as cobots

We handle the OT layer of smart factory and AI factory projects

From control panel design and build to PLC integration and a standard data API, we handle the field side so you can focus on your platform and services.

PAIN POINTS

If your projects keep stalling in the field

- ✓ Every site has different PLC models and protocols
- ✓ Installing and maintaining gateways is a burden
- ✓ It is hard to take on electrical safety for panel work
- ✓ Data formats differ by machine and are hard to organize
- ✓ You lack staff for installation and commissioning
- ✓ You have a platform but need manufacturing sites

DONG-IN ENSIS

Field data · OT

- Panel design · build · install · commissioning
- PLC · drive · sensor integration
- Encrypted transfer · DataHub (cloud / on-prem)
- Standard data model + API

IT PARTNER

Platform · services

- MES · ERP · EAM integration
- AI analytics · predictive models
- Digital twin · dashboards
- Customer-facing platform operation

MODEL 01

OT delivery

We handle control panels, PLCs and on-site installation in your project.

- Panel design · build · install
- PLC · sensors
- Commissioning · O&M

MODEL 02

Data integration

Equipment data collected by the panel is delivered to your platform through a standard API.

- Standard data model
- REST API · WebSocket
- MES · AI · digital twin

MODEL 03

Joint proposals · sales

We find manufacturing customers and government programs together and propose jointly.

- Joint customer discovery
- Support programs
- Demo equipment

Different equipment, the same data structure

- 01

Grouped by equipment
site_id and equipment_id identify which machine at which site.
- 02

One format for every sensor
Temperature, current or vibration — every sensor uses the same value, unit and status format.
- 03

Status from thresholds
Each sensor has warning and alarm thresholds; if any sensor warns, the equipment status becomes warning.

REST API

Equipment and sensor lists, latest values, history

WebSocket

Real-time values and status changes

CSV

Export for testing and analysis

```
// Example data (for illustration)
{
  "site_id": "PLANT-01",
  "equipment_id": "PUMP-01",
  "timestamp": "2026-10-07T09:30:00+09:00",
  "status": "warning",
  "sensors": [
    { "id": "bearing_temp", "value": 48.2, "unit": "°C", "status": "normal" },
    { "id": "motor_current", "value": 31.6, "unit": "A", "status": "normal" },
    { "id": "vibration", "value": 4.8, "unit": "mm/s", "status": "warning" }
  ]
}
```

How we work
From inquiry to joint projects

1 Inquiry

Share your solution and target project

2 Tech meeting

Agree scope and roles (NDA if needed)

3 Demo · test

Verify integration with demo equipment and test data

4 Joint project

Customer deployment, joint proposal and delivery

Resources
donginensis.com
IT partner page

PDF

Data integration overview
Architecture · data model · security

JSON

Sample data (real-time)
1 machine · 3 sensors · thresholds

CSV

Sample data (time series)
24 h · 10-min interval · warning

SEC

Security overview
Network · encryption · access

(주)동인엔시스

**Controlling equipment,
connecting data.**



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